

Scleroderma — Treatment by Organ

LEFT ZONE (55%) THE TREATMENT WORKSTATIONS

Station 1 - ILD

Station 2 - PAH

Station 3 - SRC

Station 4 - RAYNAUD/VASCULAR

Station 5 - GI

Station 6 - SKIN/MSK

THE DRAGON TAMER'S WORKSHOP

Frame 1 - MORPHEA

Frame 2 - MCTD

Frame 3 - EOSINOPHILIC FASCIITIS

Raynaud	✓	✗	✓	✗
Internal organs	✓	✗	✓	✗
Depth	DERMIS	DERMIS	DERMIS	FASCIA
Antibodies	SSc	NONE	U1-RNP	NONE
Eosinophilia	✗	✗	✗	✓
Steroids	☠	~	✓	✓✓

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RIGHT ZONE (45%) THE COMPARISON GALLERY

Frame 1 - MORPHEA

Frame 2 - MCTD

Frame 3 - EOSINOPHILIC FASCIITIS

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Steroids	☠	~	✓	✓✓

1. No cure / organ-based therapy

WHAT YOU SEE

NO CURE — ORGAN-BASED, AVOID STEROIDS banner top center

WHAT IT ENCODES

There is no cure for systemic sclerosis; management is ORGAN-BASED, targeting each affected system separately. The mantra: ACEi for the kidney (renal crisis), PPI for the gut (reflux), mycophenolate/cyclophosphamide for the lung (ILD), vasodilators for the vessels (Raynaud/PAH). Established fibrosis is not reversible, so therapy halts progression and treats complications.

BOARD TRAP

Picking a single 'disease-modifying' drug to treat scleroderma globally — WRONG. No one agent reverses SSc; you treat organ by organ. Discriminator: the right answer always names the therapy matched to the specific organ complication in the stem (e.g., captopril for renal crisis), not a blanket immunosuppressant.

2. Avoid steroids (renal crisis risk)

WHAT YOU SEE

AVOID STEROIDS warning under header

WHAT IT ENCODES

High-dose corticosteroids (≥15 mg/day prednisone) are AVOIDED in SSc, especially early diffuse disease, because they precipitate scleroderma renal crisis. This is a major exception to the rheumatology rule that steroids treat most autoimmune flares. If unavoidable, use the lowest dose with close BP and creatinine monitoring.

BOARD TRAP

Choosing high-dose prednisone for tightening diffuse skin or a small pericardial effusion — WRONG. Steroids trigger renal crisis in diffuse SSc. Discriminator: when a stem offers steroids for SSc skin/serositis, the correct move is a steroid-sparing agent (methotrexate, mycophenolate) — reserve any steroid only for clear inflammatory overlap and at low dose.

3. ILD: MMF first-line

WHAT YOU SEE

Station 1 ILD: MMF 1st line vial, CYC

WHAT IT ENCODES

Mycophenolate mofetil (MMF, ~target 1-1.5 g twice daily) is now first-line for SSc-ILD, having matched cyclophosphamide efficacy in the Scleroderma Lung Study II with far better tolerability and less toxicity. Cyclophosphamide is reserved for severe/rapidly progressive disease. Treat based on objective FVC/DLCO decline and HRCT extent.

BOARD TRAP

Defaulting to cyclophosphamide as first-line for SSc-ILD — WRONG. MMF is preferred first-line (equal efficacy, fewer adverse effects like hemorrhagic cystitis, marrow suppression, and malignancy). Discriminator: reserve cyclophosphamide for severe or refractory ILD; choose MMF for typical progressive SSc-ILD.

4. ILD: nintedanib / tocilizumab

WHAT YOU SEE

Tocilizumab + antifibrotic vials in ILD station

WHAT IT ENCODES

Add-on options for SSc-ILD: nintedanib (a tyrosine-kinase antifibrotic, FDA-approved for SSc-ILD, slows the rate of FVC decline) and tocilizumab (anti-IL-6R, approved to preserve lung function in early inflammatory SSc-ILD with elevated inflammatory markers). These complement, rather than replace, MMF.

BOARD TRAP

Expecting nintedanib to reverse established lung fibrosis — WRONG. Antifibrotics SLOW the rate of FVC decline; they do not restore lost lung. Discriminator: nintedanib's endpoint in trials was reduced annual FVC decline, not improvement — set expectations and combine with immunosuppression in inflammatory disease.

5. PAH: vasodilator triple class

WHAT YOU SEE

Station 2 PAH: bosentan (ERA), sildenafil (PDE-5i), prostacyclin

WHAT IT ENCODES

SSc-PAH (Group 1) is treated with PAH-specific vasodilators across three pathways: endothelin receptor antagonists (bosentan, ambrisentan, macitentan), PDE-5 inhibitors (sildenafil, tadalafil) or the sGC stimulator riociguat, and prostacyclin analogs (epoprostenol, treprostinil, oral selexipag). Early combination therapy improves outcomes; epoprostenol improves survival in severe disease.

BOARD TRAP

Treating SSc-PAH with high-dose calcium channel blockers (the Raynaud drug) as you would idiopathic vasoreactive PAH — WRONG. SSc-PAH patients are almost never CCB-responders. Discriminator: idiopathic PAH with a positive acute vasoreactivity test may use high-dose CCBs, but SSc-PAH requires the three-class PAH-specific agents above. NEVER combine riociguat with a PDE-5 inhibitor (severe hypotension).

6. PAH: avoid digoxin/diuretic pitfalls

WHAT YOU SEE

Diuretics / digoxin AVOID icons under PAH bench

WHAT IT ENCODES

Supportive PAH care: cautious diuretics for RV volume overload (the failing RV is preload-dependent — over-diuresis drops cardiac output), supplemental O2 for hypoxemia, and treat the right heart. Aggressive diuresis and unnecessary negative inotropy/AV-nodal slowing can precipitate RV failure and hypotension.

BOARD TRAP

Aggressively diuretizing a hypotensive SSc-PAH patient with elevated JVP to 'unload' the heart — WRONG. The pressure-loaded RV depends on preload; over-diuresis collapses cardiac output. Discriminator: in right-heart-dominant PAH, optimize volume gently and start PAH-specific vasodilators — do not treat it like left-heart failure with brisk diuresis.

7. SRC: ACE inhibitor (lifesaving)

WHAT YOU SEE

Station 3 SRC: ACEi vial, do-not-stop warning

WHAT IT ENCODES

Scleroderma renal crisis is treated with immediate, rapidly titrated short-acting ACE inhibitors (captopril, e.g. 6.25-12.5 mg uptitrated every few hours to control BP). Continue the ACEi even as creatinine rises and even if temporary dialysis is required, because renal recovery occurs in many patients over weeks to months. ACEi transformed SRC from near-uniformly fatal to survivable.

BOARD TRAP

Substituting an ARB for the ACE inhibitor, or stopping the ACEi when creatinine climbs — WRONG. ACEi (not ARB) is the proven first-line in SRC, and you do not stop it for a rising creatinine. Discriminator: ARBs lack the same evidence base in SRC; the standard answer is captopril, continued through the expected creatinine bump, with dialysis as a bridge if needed.

8. Raynaud: CCB first-line

WHAT YOU SEE

Station 4 Raynaud/Vascular: CCB (nifedipine), warm gloves

WHAT IT ENCODES

Raynaud management starts with non-drug measures (keep core/hands warm, avoid cold/nicotine/sympathomimetics/beta-blockers) then a dihydropyridine calcium channel blocker — nifedipine (extended-release ~30-60 mg/day) or amlodipine — as first-line pharmacotherapy. Topical nitrates are an adjunct.

BOARD TRAP

Reaching for a non-dihydropyridine or a beta-blocker, or starting a PDE-5 inhibitor before a CCB in uncomplicated Raynaud — WRONG. Dihydropyridine CCBs are first-line; beta-blockers WORSEN vasospasm. Discriminator: escalate to PDE-5 inhibitors/iloprost only for refractory disease or digital ulcers, not as the initial agent.

9. Digital ulcers: PDE-5i / iloprost / bosentan

WHAT YOU SEE

PDE-5i, iloprost, prostacyclin for refractory digits

WHAT IT ENCODES

For refractory Raynaud and active digital ulcers: add PDE-5 inhibitors (sildenafil/tadalafil) and IV prostacyclin (iloprost) for healing; bosentan (endothelin antagonist) is used specifically to PREVENT new digital ulcers in patients with recurrent ulceration. Surgical/chemical digital sympathectomy for critical ischemia.

BOARD TRAP

Choosing bosentan to heal an existing acute digital ulcer — WRONG. Bosentan's evidence is for PREVENTING new ulcers, not accelerating healing of current ones. Discriminator: to heal an active ulcer use IV iloprost and/or PDE-5 inhibitors; use bosentan to reduce the number of future ulcers in recurrent disease.

10. GI: PPI + prokinetics

WHAT YOU SEE

Station 5 GI: PPI, prokinetic, dysmotility icons

WHAT IT ENCODES

GI dysmotility is nearly universal (the esophagus is the most commonly involved internal organ). Treat reflux/esophagitis with PPIs (often high-dose, sometimes twice daily) to prevent strictures and Barrett's; add prokinetics (metoclopramide, domperidone) for dysmotility, gastroparesis, and pseudo-obstruction. Lifestyle: small meals, head-of-bed elevation.

BOARD TRAP

Treating severe SSc reflux with lifestyle and intermittent antacids alone — WRONG. SSc esophageal disease usually needs chronic, often high-dose, PPI therapy. Discriminator: undertreated reflux leads to peptic strictures and Barrett esophagus; the correct answer escalates acid suppression and adds prokinetics rather than relying on PRN antacids.

11. GI: rotating antibiotics for SIBO

WHAT YOU SEE

SIBO rotating antibiotics icon at GI station

WHAT IT ENCODES

Small intestinal bacterial overgrowth (SIBO) results from small-bowel hypomotility, causing bloating, diarrhea, and malabsorption with B12/fat-soluble-vitamin deficiency and weight loss. Treat with cyclical/rotating courses of antibiotics (e.g., rifaximin, metronidazole, ciprofloxacin) to suppress overgrowth, plus nutritional support.

BOARD TRAP

Diagnosing SSc diarrhea/weight loss as inflammatory bowel disease and starting immunosuppression — WRONG. Malabsorptive diarrhea in SSc is usually dysmotility-driven SIBO. Discriminator: SIBO responds to rotating antibiotics and improves with prokinetics; an abnormal hydrogen breath test and the SSc dysmotility context point away from IBD.

12. Skin/MSK: MTX, HSCT for severe

WHAT YOU SEE

Station 6 Skin/MSK: MTX, exercise bands, HSCT for severe diffuse

WHAT IT ENCODES

For active diffuse skin disease, methotrexate or mycophenolate are the steroid-sparing agents of choice; physical/occupational therapy preserves hand function and prevents contractures. Autologous hematopoietic stem cell transplant (HSCT) is reserved for severe, rapidly progressive early diffuse disease with organ involvement — it improves survival but carries significant upfront treatment-related mortality.

BOARD TRAP

Offering HSCT to a stable limited-SSc patient, or reaching for steroids for skin tightening — WRONG. HSCT is only for severe progressive EARLY DIFFUSE disease, and steroids precipitate renal crisis. Discriminator: match intensity to severity — methotrexate/mycophenolate for typical diffuse skin, HSCT only for high-risk rapidly progressive diffuse disease in selected centers.

13. MCTD comparison (anti-U1-RNP)

WHAT YOU SEE

Comparison Gallery Frame 2 MCTD wolf, anti-U1-RNP, responds to steroids

WHAT IT ENCODES

Mixed connective tissue disease (MCTD) is an overlap of SLE, systemic sclerosis, and polymyositis features (Raynaud, puffy hands, arthritis, myositis) defined by HIGH-TITER anti-U1-RNP antibody. Unlike SSc, MCTD is generally STEROID-RESPONSIVE and has a better prognosis, though PAH is its leading cause of death.

BOARD TRAP

Withholding steroids from an anti-U1-RNP MCTD patient because 'steroids cause renal crisis in scleroderma' — WRONG. MCTD does respond to and is treated with steroids. Discriminator: high-titer anti-U1-RNP = MCTD (steroid-responsive overlap); anti-Scl-70/centromere/RNA-pol III = systemic sclerosis (avoid steroids). Match antibody to whether steroids help or harm.

14. Eosinophilic fasciitis (mimic)

WHAT YOU SEE

Frame 3 eosinophilic fasciitis, groove sign, fascia not dermis, MTX/steroids

WHAT IT ENCODES

Eosinophilic fasciitis is a scleroderma MIMIC affecting the FASCIA (not dermis), typically after vigorous exercise, with symmetric limb induration, peau d'orange skin, the 'groove sign' (depressions along veins), and peripheral EOSINOPHILIA. It SPARES the fingers and has no Raynaud or ANA. Full-thickness skin-to-muscle biopsy confirms; it RESPONDS WELL to corticosteroids (± methotrexate).

BOARD TRAP

Diagnosing systemic sclerosis in a patient with arm induration after exercise and withholding steroids — WRONG. Eosinophilic fasciitis lacks Raynaud, lacks ANA, spares the digits, has eosinophilia, and is STEROID-responsive. Discriminator: no Raynaud + no nailfold changes + finger sparing + peripheral eosinophilia + groove sign = eosinophilic fasciitis, where steroids are the correct treatment (the opposite of SSc).